



Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

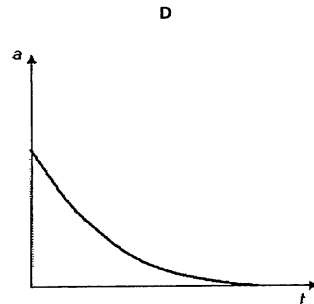
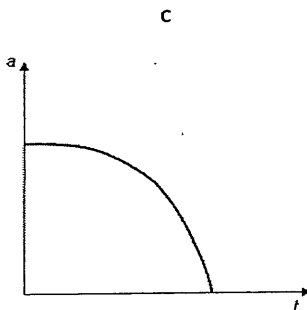
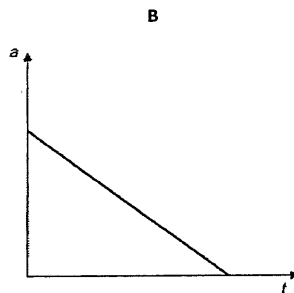
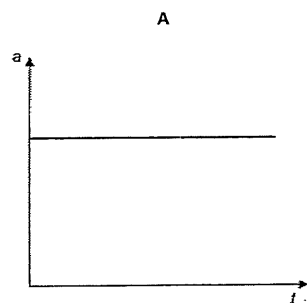
1. [DHS/Prelims 2014/P1/5]

What is meant by the weight of an object?

- A the gravitational field acting on the object
- B the gravitational force acting on the object
- C the mass of the object multiplied by gravity
- D the object's mass multiplied by its acceleration

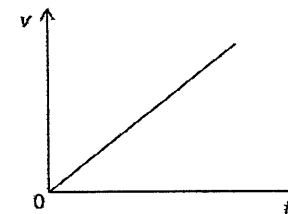
2. [CJC/Prelims 2014/P1/5]

A metal sphere is held just below the surface of a deep tank of liquid and released. Which graph best illustrates how the acceleration a varies with time t after release?

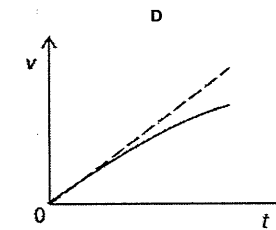
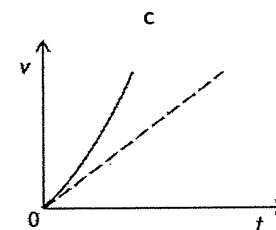
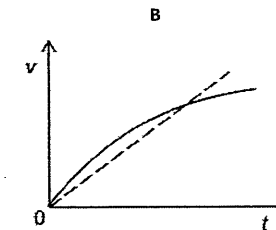
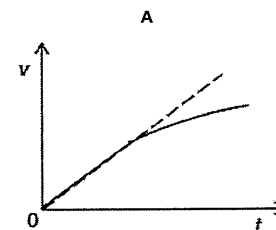


3. [DHS/Prelims 2014/P1/3]

A body falls from rest in a vacuum near the Earth's surface. The variation with time t of its speed v is shown below.



Which graph shows the variation with time t of the speed v of the same ball falling in air at the same place on Earth?



4. [HCI/Prelims 2014/P1/3]

A raindrop, in still air, falls at a constant vertical terminal velocity of 1.8 m s^{-1} . If the raindrop were to fall through air in which a horizontal wind of velocity 1.4 m s^{-1} is blowing, calculate the magnitude of the resultant velocity of the raindrop.

- A 0.40 m s^{-1} B 1.8 m s^{-1} C 2.3 m s^{-1} D 3.2 m s^{-1}

5. [HCI/Prelims 2014/P1/5]

A bullet is aimed directly and horizontally at the centre of a target which is 100 m away. If the bullet is then fired at a speed of 200 m s^{-1} , what is the distance the bullet will land from the centre of the target?

- A 0 cm B 12 cm C 123 cm D 196 cm

6. [IJC/Prelims 2014/P1/4]

A projectile is launched at an angle of 30° to the horizontal. Its initial kinetic energy is E . If air resistance is negligible, what is the projectile's kinetic energy at the highest point of motion?

- A zero B $0.50 E$ C $0.75 E$ D E

7. [MJC/Prelims 2014/P1/3]

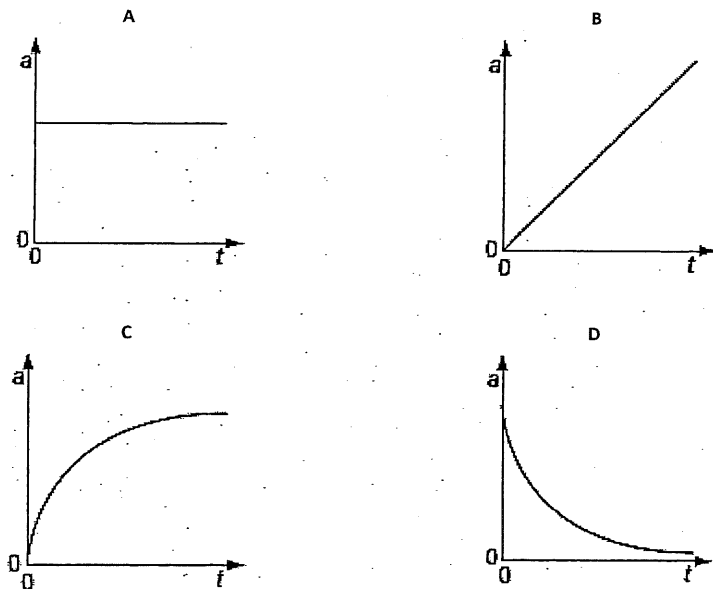
A stone is thrown at an angle and follows a parabolic path. The highest point reached is A and the point just before it reaches the ground is B. Assuming that air resistance is negligible, the vertical acceleration of the stone is

- A zero at A.
B greatest at A.
C greatest at B.
D the same at A and B.

8. [NYJC/Prelims 2014/P1/3]

A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time t of the acceleration a of the ball as it falls, assuming that the effects of air resistance are appreciable?



9. [NYJC/Prelims 2014/P1/4]

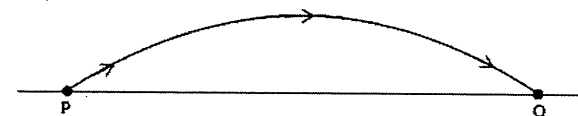
A hot air balloon carrying a passenger is descending at a constant velocity of 20.0 m s^{-1} . The passenger throws a stone horizontally at 15.0 m s^{-1} and 2.00 s later, the rock strikes the ground.

What is the speed at which the rock strikes the ground?

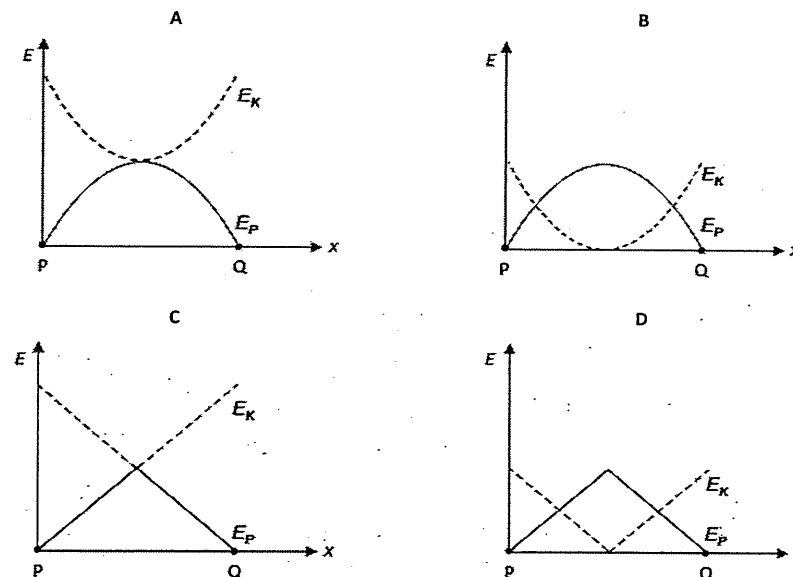
- A 15.0 m s^{-1} B 30.0 m s^{-1} C 42.4 m s^{-1} D 59.6 m s^{-1}

10. [PJC/Prelims 2014/P1/4]

In the absence of air resistance, a stone is thrown from point P and follows a parabolic path before it lands at point Q.

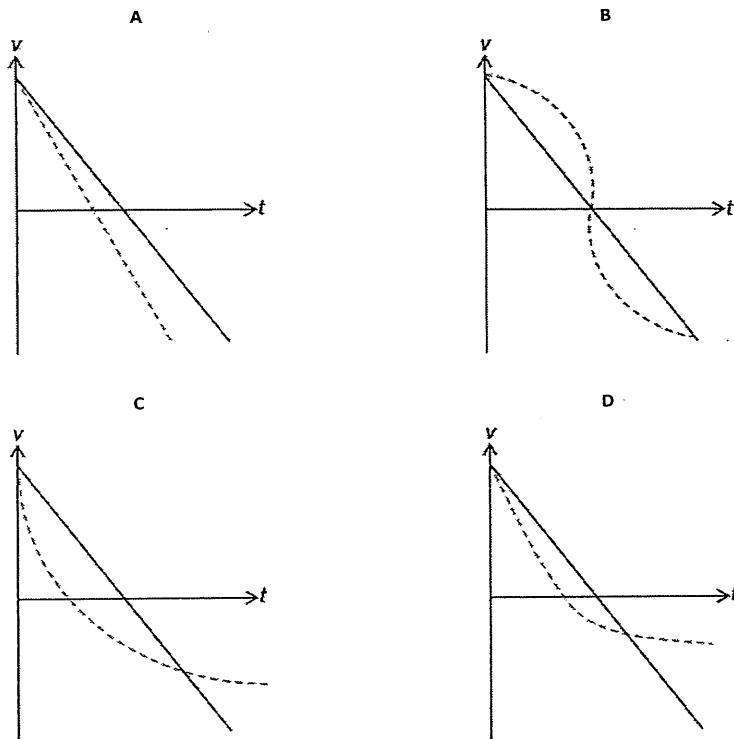


Which graph best shows the variation with horizontal distance x of the kinetic energy E_K and potential energy E_P of the ball?



11. [SAJC/Prelims 2014/P1/3]

A ball is thrown vertically upward. It reaches the highest point and comes back down to the original starting point. Which is the correct velocity-time graph for the ball without air resistance (solid line) and with air resistance (dotted line)?



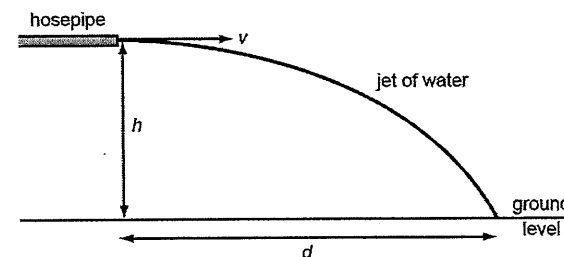
12. [SAJC/Prelims 2014/P1/4]

A ball X of mass m is projected horizontally with velocity v from a certain height. At the same time, another ball Y of mass $2m$ is fired from an angle above the horizontal with the same velocity from the same point. When will X hit the floor compared to Y. (Neglect air resistance.)

- A X and Y will hit the floor at the same time at the same speed and kinetic energy.
- B X will hit the floor before Y at different speed but same kinetic energy.
- C X will hit the floor before Y at same speed but different kinetic energy.
- D X will hit the floor after Y at the same speed and kinetic energy.

13. [SAJC/Prelims 2014/P1/5]

A hosepipe is fixed as shown.



The jet of water emerges with a horizontal velocity v . The hosepipe is fixed at a height h above the ground. The water jet hits the floor at a horizontal distance d from the nozzle tip. The gravitational field strength is g . What is the expression for distance d ? (Ignore air resistance.)

- A $d = \frac{vg}{2h}$
- B $d = \frac{2vh}{g}$
- C $d = v\sqrt{\frac{g}{2h}}$
- D $d = v\sqrt{\frac{2h}{g}}$

14. [SRJC/Prelims 2014/P1/4]

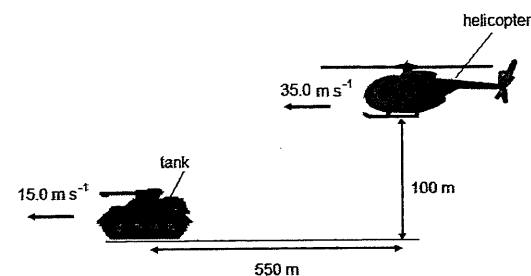
A wingsuit flyer jumps off a 300 m tall building with a constant speed of 100 m s^{-1} at an angle of 30° below the horizontal. After falling for 3.0 s, he drops an object. Air resistance is negligible.

What is the time taken for the object to fall onto the ground from the time he drops it?

- A 1.59 s
- B 2.02 s
- C 2.42 s
- D 3.00 s

15. [TJC/Prelims 2014/P1/3]

In the figure below, a helicopter is flying with a constant horizontal speed of 35.0 m s^{-1} and is 100 m above the ground. A tank is traveling in the same direction with a constant horizontal speed 15.0 m s^{-1} and is 550 m away from the helicopter.



Assume that at $t = 0$ s, the helicopter and the tank are at the above position.

How many seconds later should the helicopter drop a bomb so that it will hit the tank? Ignore air resistance.

- A 4.52 s
- B 10.2 s
- C 23.0 s
- D 27.5 s

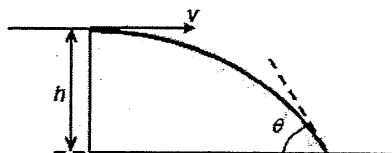
16. [VJC/Prelims 2014/P1/3]

A rock is thrown with an initial kinetic energy of 15 J at an angle of 30° to the horizontal. What is its kinetic energy at the highest point of its trajectory?

- A 0 J B 3.8 J C 7.5 J D 11 J

17. [YJC/Prelims 2014/P1/3]

The diagram shows the path of a projectile fired with a horizontal velocity v from the top of a cliff of height h .

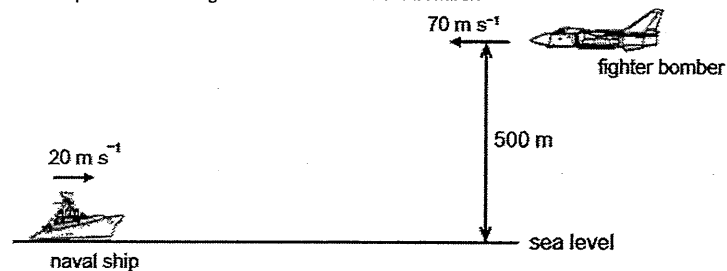


Which of the following values of v and h will give the greatest value of the angle θ ?

	$v / \text{m s}^{-1}$	h / m
A	10	20
B	10	80
C	30	10
D	50	10

18. [YJC/Prelims 2014/P1/3]

A fighter bomber is travelling at 70 m s^{-1} horizontally at a height of 500 m above sea level. It is on a bombing run to its target, a naval ship which is moving at 20 m s^{-1} towards the bomber.



If the pilot wants to score a direct hit, he should release the bomb

- A when the plane is 260 m away from the naval ship horizontally.
 B when the plane is 470 m away from the naval ship horizontally.
 C when the plane is 710 m away from the naval ship horizontally.
 D when the plane is 910 m away from the naval ship horizontally.

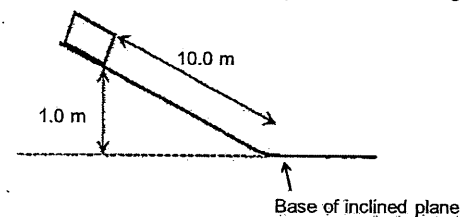
19. [AJC/Prelims 2014/P1/10]

A motor driving a pump raises 0.10 m^3 of water through a vertical height of 5.0 m in a time of 10 minutes. If the efficiency of the pump is 60%, what is the power generated by the motor? (Density of water is 1000 kg m^{-3})

- A 4.9 W B 8.2 W C 14 W D 82 W

20. [CJC/Prelims 2014/P1/9]

A body of mass 1.0 kg initially at rest slides down an inclined plane that is 1.0 m high and 10.0 m long as shown.

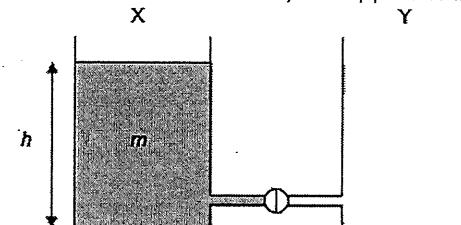


The body experiences a constant resistive force of 0.50 N while travelling on the slope. What is the kinetic energy of the body at the base of the plane?

- A 4.8 J B 9.3 J C 10 J D 15 J

21. [DHS/Prelims 2014/P1/9]

The diagram shows two identical vessels X and Y connected by a short pipe with a tap.



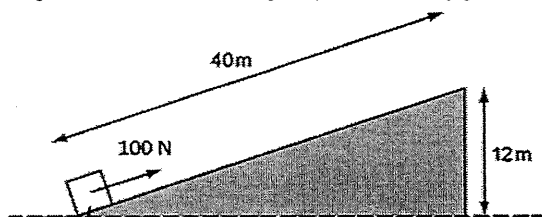
Initially, X is filled with water of mass m to a depth h , and Y is empty. When the tap is opened, water flows from X to Y until the depths of water in both vessels are equal.

How much potential energy is lost by the water during this process?

- A 0 B $\frac{mgh}{4}$ C $\frac{mgh}{2}$ D mgh

22. [H1/Prelims 2014/P1/9]

A constant force of 100 N, parallel to a rough inclined plane, moves a body of mass 20.0 kg at constant speed of 5.0 m s^{-1} through a distance of 40 m along the plane. The body gains a height of 12 m.

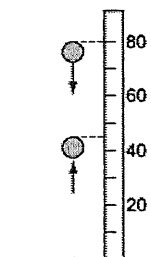


What is the magnitude of work done by friction that is dissipated as heat?

- A 1.6 kJ B 2.4 kJ C 3.7 kJ D 4.0 kJ

23. [JJC/Prelims 2014/P1/10]

A solid rubber ball has a diameter of 8.0 cm. It is released from rest with the top of the ball 80 cm above a horizontal surface. It falls vertically and then bounces back up so that the maximum height reached by the top of the ball is 45 cm.



The kinetic energy of the ball is 0.75 J just before it strikes the surface.

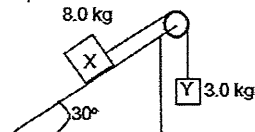
What is its kinetic energy just after it leaves the surface?

- A 0.36 J B 0.39 J C 0.40 J D 0.42 J

24. [JJC/Prelims 2014/P1/9]

A system of two bodies X and Y weighing 8.0 kg and 3.0 kg respectively are connected by a light cord that is passed over a light-free running pulley, as shown below.

Starting from rest, X moves down a smooth plane inclined at 30° to the horizontal.



When X has travelled 2.0 m along the plane, what is the total kinetic energy of the system?

- A 2.0 J B 20 J C 77 J D 98 J

25. [JJC/Prelims 2014/P1/10]

A pump draws a volume X of water up a vertical height h from a slow-moving river and discharges it through a nozzle with speed v . The entire process takes place in time t . The density of water is ρ .

What is the effective power of the pump?

- A $\frac{\rho Xgh}{t}$ B $\frac{\rho Xv^2}{2t}$ C $\frac{\rho Xgh + 2\rho Xv^2}{2t}$ D $\frac{2\rho Xgh + \rho Xv^2}{2t}$

26. [NYJC/Prelims 2014/P1/9]

A particle is dropped from rest at a height h above the surface of a viscous liquid column of height 0.80 m. It attains a speed of 7.5 m s^{-1} after it has penetrated 0.60 m of the viscous liquid column.

If the mass of the particle is 0.80 kg and the viscous liquid offers a constant retarding force of 120 N, what is the height, h , from which the particle is released?

- A 2.3 m B 3.5 m C 11.4 m D 15.7 m

27. [NYJC/Prelims 2014/P1/10]

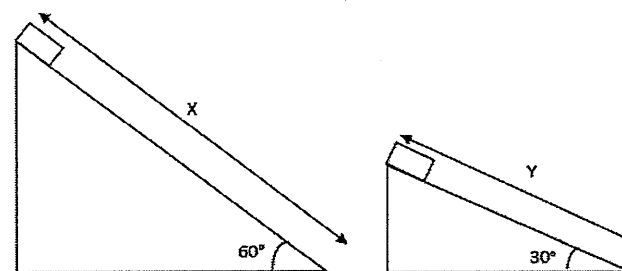
An object of mass 400 kg is lifted through a vertical height of 1200 m in 2.0 minutes by an electric motor.

What is the amount of electrical power needed if the overall efficiency of the system is 80%?

- A 3.1 kW B 4.9 kW C 49 kW D 2900 kW

28. [PJC/Prelims 2014/P1/9]

Two identical blocks are released from rest from the top of two ramps as shown.



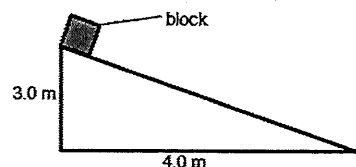
The ratio of the length X to the length Y is 2.0.

Assuming that there is no friction, what is the ratio of the blocks' speeds at the bottom of the ramps?

- A 1.1 B 1.9 C 2.0 D 3.5

29. [RVHS/Prelims 2014/P1/9]

A block of mass 2.0 kg is released from rest at the top of an inclined plane as shown.



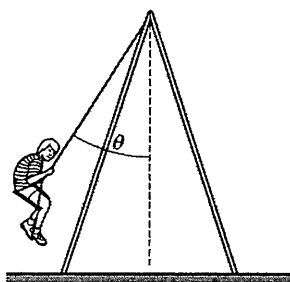
The block experiences a constant resistive force of 4.0 N throughout its motion.

What is the speed of the body at the bottom of the slope?

- A 6.2 m s^{-1} B 6.5 m s^{-1} C 7.7 m s^{-1} D 8.9 m s^{-1}

30. [SAJC/Prelims 2014/P1/10]

A child of mass 50 kg is on a swing which is suspended by 4.0 m ropes from a rigid support. The horizontal speed of the swing as it passes through the lowest point is 3.0 m s^{-1} .

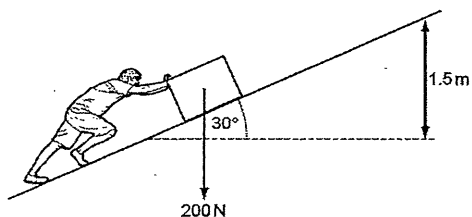


What is the angle θ that the ropes make with the vertical when the swing is at its highest point?

- A 28° B 40° C 42° D 62°

31. [TJC/Prelims 2014/P1/11]

A person pushes a box of weight 200 N so that it moves at a steady speed along a ramp, through a height of 1.5 m. The ramp makes an angle of 30° with the ground. The frictional force on the box is 150 N while the box is moving.



What is the work done by the person?

- A 170 J B 300 J C 450 J D 750 J

32. [YJC/Prelims 2014/P1/9]

A small electric motor is used to raise a weight of 10 N at constant speed through a vertical height of 100 cm in 5.0 s. The efficiency of the motor is 25 %.

What is the electrical power supplied to the motor?

- A 2.0 W B 4.0 W C 6.0 W D 8.0 W

II. Structured Questions

1. [ACJC/Prelims 2014/P2/7]

- a. Define acceleration. [1]
- b. Fig 1.1 shows how the vertical component of the velocity of a parachutist changes with time during the first 20 s of his jump. To avoid air turbulence caused by the aircraft, he waits a short time after jumping before pulling the cord to release his parachute.

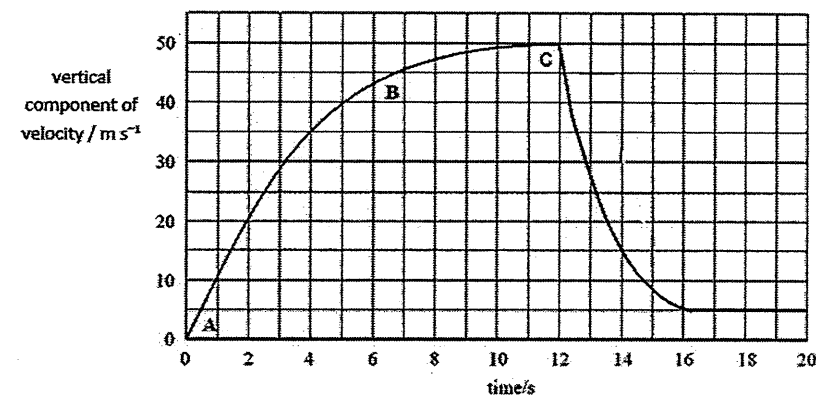


Fig. 1.1

- i. Using Fig 1.1, determine the total vertical distance fallen by the parachutist in the first 11 s of the jump. Show your method clearly. [3]
- ii. Regions A (0 to 2 s), B (2 to 11 s) and C (11 to 12 s) of the graph show the speed before the parachute was opened. With reference to the forces acting on the parachutist, explain why the graph has this shape in the region marked – A, B and C. [6]
- iii. Determine the acceleration of the parachutist at 6.0 s. [2]
- iv. Complete Fig 1.2 below (including the blank column with a suitable ratio) and hence deduce whether the air resistance experienced by the parachutist from 3.0 s to 9.0 s is directly proportional to the square of the velocity of the parachutist. Take the acceleration of free-fall as 10.0 m s^{-2} . [4]

t / s	$v / \text{m s}^{-1}$	$a / \text{m s}^{-2}$	$(g - a) / \text{m s}^{-2}$	
3.0	28.5	6.90	3.10	
6.0	43.2			
9.0	48.0	1.20	8.80	

Fig. 1.2

- c. i. Show that the maximum deceleration of the parachutist when the parachute opened at time 12 s is about 30 m s^{-2} . [1]
- ii. Hence draw a graph on Fig 1.3 showing the variation with time t of the acceleration a of the parachutist during the first 18 s of the jump. [3]
- (No further calculations required.)

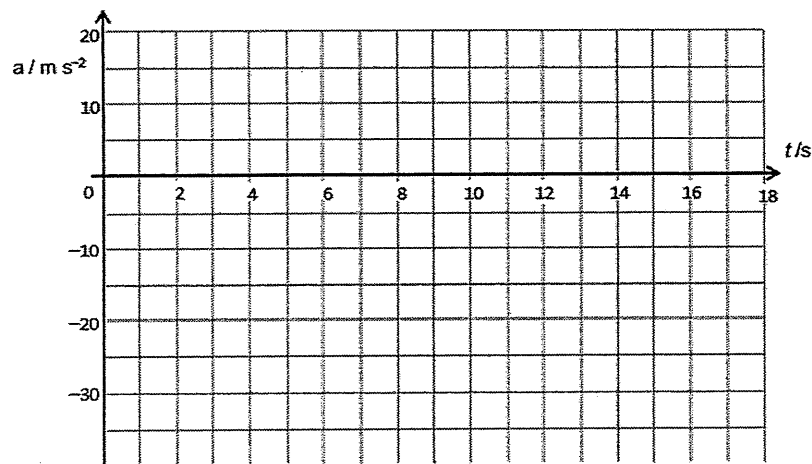


Fig. 1.3

[Total: 20]

2. [AJC/Prelims 2014/P2/1]

- a. A body has an initial velocity u and an acceleration a . After a time t , the body moved a distance s and has a final velocity v . One of its equations of motion is

$$s = ut + \frac{1}{2}at^2.$$

State two conditions that must be satisfied for the above equation to be valid. [2]

- b. A hot air balloon was rising steadily at a speed of 10.0 m s^{-1} when weather conditions turned windy. A constant breeze of 3.0 m s^{-1} blew horizontally across the sky, which caused the hot air balloon to travel with a resultant velocity of v_R at an angle θ to the horizontal, as shown in Fig. 2.1 below.

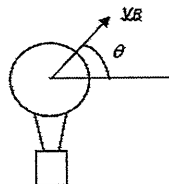


Fig. 2.1

- i. Calculate the magnitude and direction of the resultant velocity v_R . [2]
- ii. A sandbag was dropped from the balloon.
- In Fig. 2.1, sketch the path of trajectory of the sandbag as it drops from the balloon. [1]
 - Determine how far below the balloon would the sandbag be after 4.0 s. You may assume that the sandbag had not landed on the ground, the dropping of sandbag did not affect the velocity of the hot air balloon and that air resistance on the sandbag is negligible. [3]

[Total: 8]

3. [CJC/Prelims 2014/P2/1(part)]

- a. A squash ball of mass 24 g hits the wall when it reaches its maximum height of 3.2 m. It leaves a racket which is 1.7 m above the ground as seen in Fig. 3.1. The ball is incident on the wall with a horizontal velocity of 15 m s^{-1} and rebounds in a horizontal direction with a velocity of 12 m s^{-1} . The ball is in contact with the wall for 0.15 s.

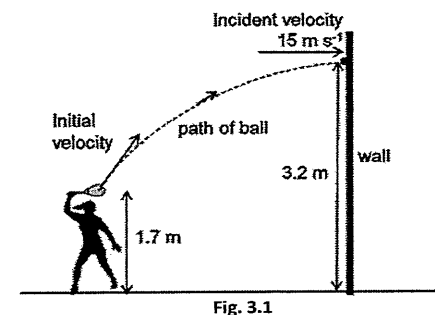


Fig. 3.1

- Calculate the initial vertical component of the ball's velocity. [2]
- Determine the average force exerted on the wall during the ball's collision with the wall. [4]
- State and explain whether the collision of the ball with the wall is elastic. [1]
- Explain why the ball does not rebound to the point from where it was hit by the racket. [2]

[Total: 9]

4. [JJC/Prelims 2014/P2/1]

A stuntman on a motorcycle plans to ride up a ramp in order to jump over a number of cars as shown in Fig. 4.1.

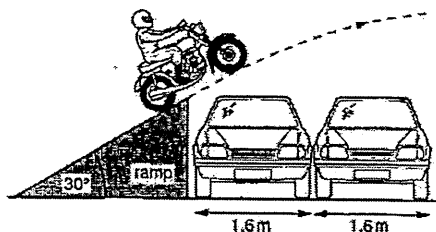
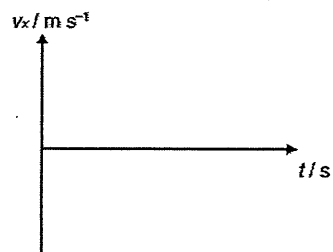


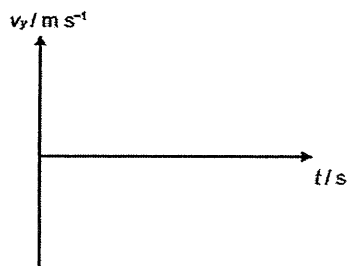
Fig. 4.1

His velocity is 14 m s^{-1} as he leaves the ramp at $t = 0 \text{ s}$. Assume that air resistance is negligible throughout the question.

- Suggest why the ramp cannot be frictionless in order for him to travel up the ramp. [1]
- The cars are each of width 1.6 m and the same height as the ramp. Estimate the maximum number of cars which he can jump over. [3]
- On the axes below, sketch the variation with time t during his flight in air for
 - the horizontal component of his velocity, v_x [1]



- the vertical component of his velocity, v_y [1]



[Total: 6]

5. [NJC/Prelims 2014/P2/2]

- State the conditions necessary for the equations of motion, $v = u + at$ and $v^2 = u^2 + 2as$, to be applicable. [2]
- An elevator starts at rest on the ninth floor. At $t = 0$, a passenger pushes a button to go to another floor. Fig 5.1 shows the acceleration, a_y , of the elevator as a function of time.

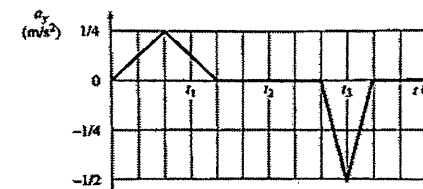


Fig. 5.1

- State and explain whether the passenger has gone to a higher or lower floor. [2]
 - Sketch a graph of velocity, v_y , of the elevator with time. [2]
- c. When baseball players throw the ball in from the outfield, they usually allow it to take one bounce before it reaches the infield, shown in path 1 in Fig. 5.2. The ball's speed is half of what it was before the bounce. Path 2 is when the ball is thrown such that it does not bounce.

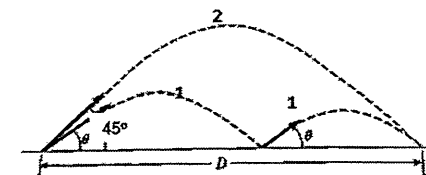


Fig. 5.2

- Determine the angle θ . You may assume that the player will always throw at the same speed. [2]
- Determine the ratio of the time interval for path 1 to the flight time for path 2. [2]

[Total: 10]

6. [NYJC/Prelims 2014/P2/1]

Astronauts plan a space expedition to Planet Newtonia to determine its acceleration of free fall.

- As part of the preliminary investigations conducted on Earth, a tennis ball is released from the top of a 12 storey building. Estimate the momentum of the tennis ball just before it hits the ground. [2]
- Upon reaching Planet Newtonia, a scientist takes measurements to determine a value for the acceleration of free fall on Planet Newtonia. A stroboscopic photograph (shown to scale) in Fig. 6.1 shows the motion of a free falling tennis ball released from rest. The strobe rate is 10 flashes per second.



Fig. 6.1

- Draw a graph on Fig. 6.2 showing how the displacement of the tennis ball varies with the square of time. [2]

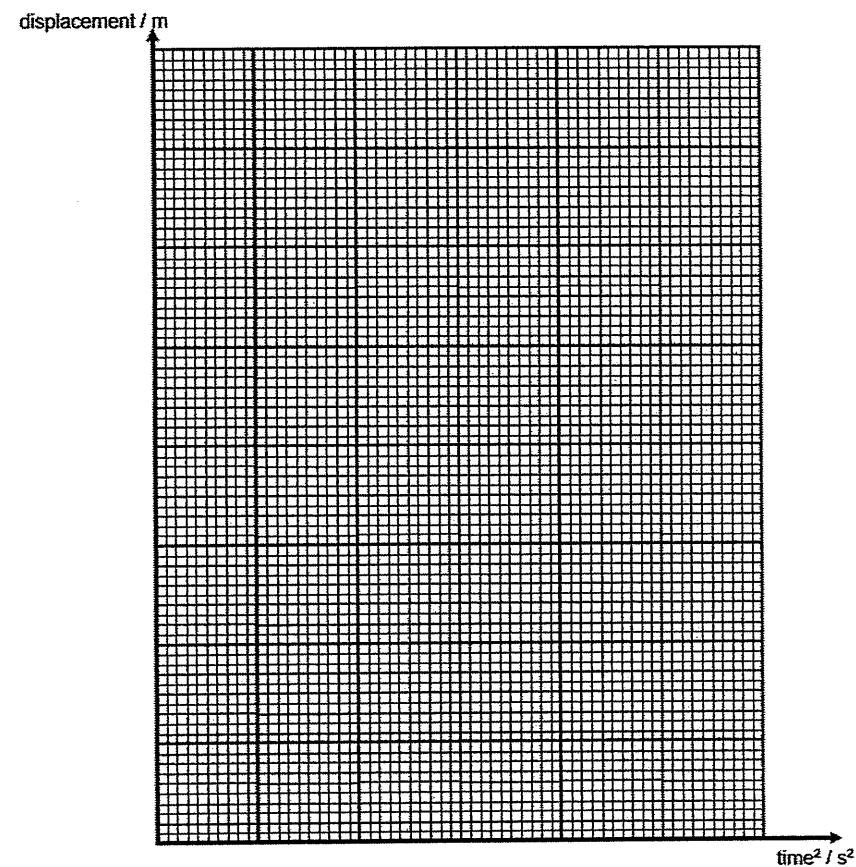


Fig. 6.2

- Hence determine the acceleration of free fall on Planet Newtonia. [2]

- c. During the space expedition on the Moon of Planet Newtonia, an astronaut is stranded on a ledge as shown in Fig. 6.3. The rescuer on the ground wants to shoot a projectile to him with a light rope attached to it. The projectile is directed at an initial angle of 55° and speed 15 m s^{-1} . The acceleration due to free fall on the Moon is 1.2 m s^{-2} .

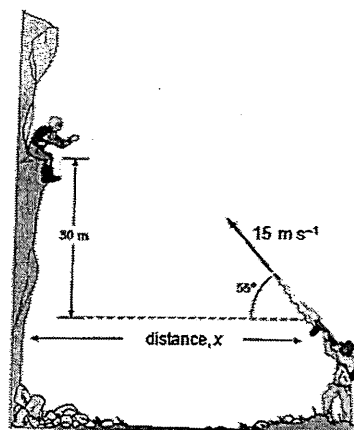


Fig. 6.3

- Determine the shortest possible time the rescuer needs to take to send the projectile to the astronaut. [2]
- Hence, calculate how far the rescuer should stand in order for the projectile to land on the ledge. [1]
- Suggest how would your answer in c. i. would change if the situation occurred on Planet Newtonia instead. [1]

[Total: 10]

7. [PJC/Prelims 2014/P2/1]

A ball is thrown with a velocity of 25 m s^{-1} vertically upwards from ground level. Air resistance is not negligible. The variation with time t of the vertical velocity v of the ball is shown in Fig. 7.1.

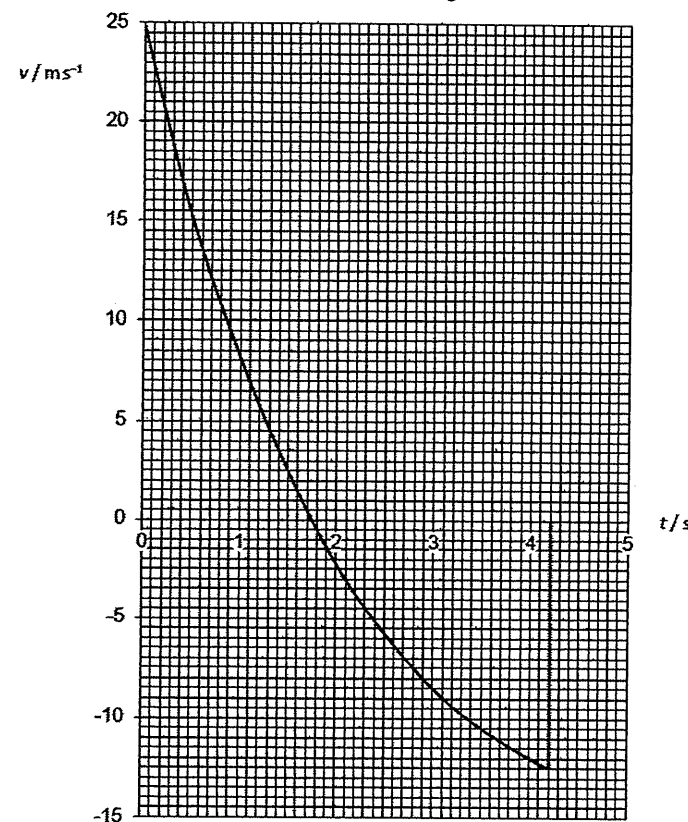


Fig. 7.1

- Use Fig. 7.1 to explain how it may be deduced that air resistance varies with speed. [2]
- The mass of the ball is 350 g and that the maximum height reached by the ball is 19 m. Use Fig. 7.1 to determine the ratio

$$\frac{\text{energy lost from the ball due to air resistance during the ball's upward motion}}{\text{energy lost from the ball due to air resistance during the ball's downward motion}}$$
 [3]
- State the time at which the acceleration is g . [1]
 - On Fig. 7.1, sketch the graph to show the variation with time t of the velocity v of the ball if air resistance is negligible. [1]

[Total: 7]

8. [SAJC/Prelims 2014/P3/1]

Frankie embarks on a hot air balloon expedition. The hot air balloon is rising vertically at a constant speed of 6.0 m s^{-1} , 50 m above ground when he tries to chase away a bird with his shoe. He throws his shoe with an initial speed of 20 m s^{-1} relative to the hot air balloon, at an angle of 30° upwards with respect to the vertical as shown in Fig. 8.1. It is a miss throw and the shoe hit the ground some time later. The mass of his shoe is 200 g.

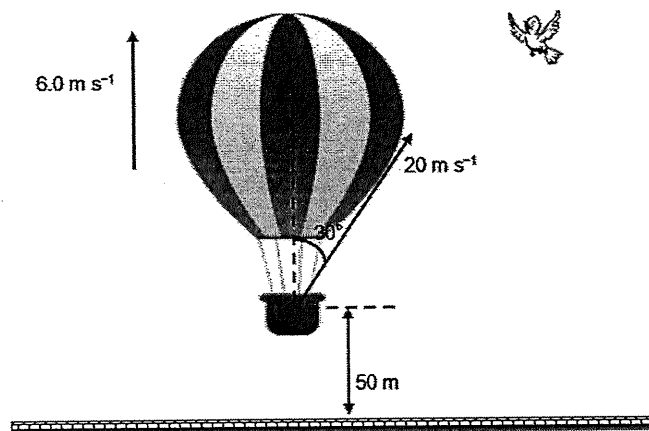


Fig. 8.1

- Calculate the maximum height above ground reached by the shoe. [2]
- Determine the time taken for the shoe to hit the ground. [2]
- The horizontal distance from the hot air balloon is x . On Fig 8.2, sketch the variation with x of the potential energy E_p and kinetic energy E_k of the shoe. Label the value(s) for the intercept(s). [3]

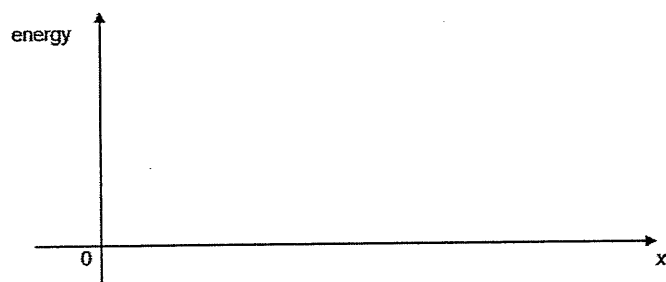


Fig. 8.2

- Taking air resistance into consideration, state and explain the effect of air resistance on the path taken by the shoe. [2]

[Total: 9]

9. [SRJC/Prelims 2014/P2/1]

A stone is thrown horizontally with a speed of 8.20 m s^{-1} from the top of a cliff into the sea. The path of the stone is shown in Fig 9.1.

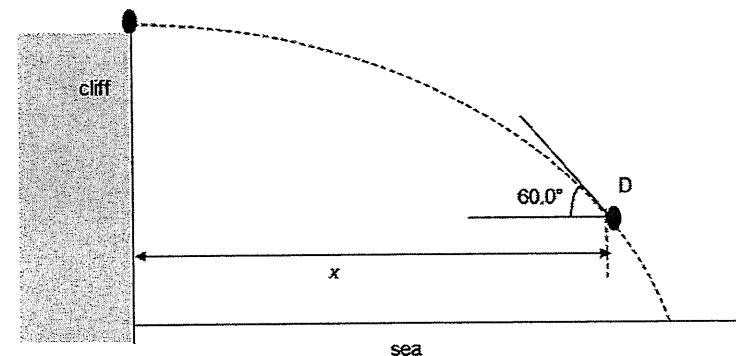


Fig. 9.1

At point D on the path of the stone, the stone is at a horizontal distance x from the cliff and is moving at an angle of 60.0° to the horizontal. Air resistance is negligible.

- For the stone at point D,
 - show that the vertical component of its velocity is 14.2 m s^{-1} . [1]
 - calculate the horizontal distance x . [2]
- Another stone is thrown vertically downwards into the sea such that it enters the sea at a high speed.
 - Describe the motion of the stone after it hits the sea. [2]
 - State for a similar stone of a heavier mass whether it will reach the same, higher or lower terminal velocity if it enters the sea with the same speed as the lighter stone. [1]

[Total: 6]

10. [YJC/Prelims 2014/P3/6]

- a. Define acceleration. [1]
- b. A 2.0 kg toy rocket is propelled vertically upwards. The graph of Fig. 10.1 shows the variation with time t of the velocity v of a toy rocket from the moment the fuel of the rocket is used up.

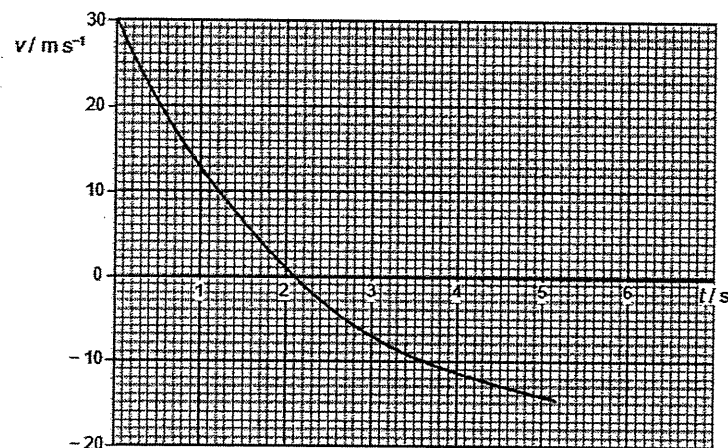


Fig. 10.1

- i. The rocket is propelled with a constant acceleration before the fuel of the rocket is used up. Using Fig. 10.1, state and explain how the propulsion force provided by the fuel varies with time before the fuel of the rocket is used up. Assume that the mass of the fuel is negligible, compared to the mass of the rocket. [3]
- ii. Using Fig. 10.1, determine the acceleration of the toy rocket at $t = 0$. [1]
- iii. Explain why the acceleration calculated in part (ii) is more than acceleration of free fall. [2]
- iv. State and explain the time at which the acceleration equals to acceleration of free fall. [2]
- v. Without further calculations, sketch the acceleration-time graph in Fig. 10.2 for the first 3 seconds of the motion after the fuel is used up. Label critical values on the graphs. [2]

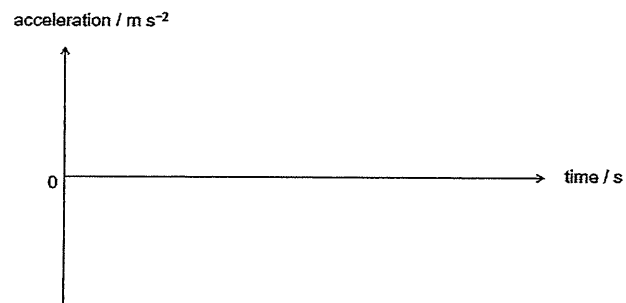


Fig. 10.2

- vi. Without further calculations, sketch the displacement-time graph in Fig. 10.3 for the first 3 seconds of the motion after the fuel is used up. Displacement is taken to be zero at the position when the fuel is used up. Label critical values on the time axes. [2]

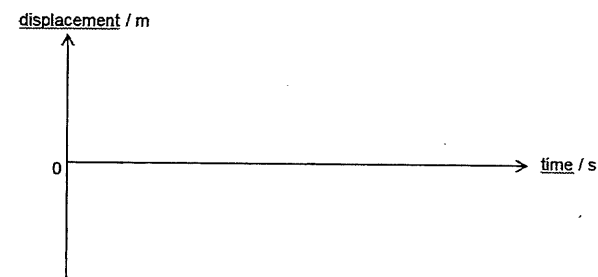


Fig. 10.3

- vii. Sketch on Fig. 10.1 the variation with time of the velocity if the rocket is replaced with one which has a larger surface area. The velocity at $t = 0$ can be assumed to remain at 30 m s^{-1} . [1]
- viii. The toy rocket will reach a speed of 22 m s^{-1} after some time before hitting a water surface and continue to move downwards with a terminal velocity of 3.0 m s^{-1} .
- Given that the velocity of the toy rocket is reduced from 22 m s^{-1} to 3.0 m s^{-1} in 3.0 ms , determine the magnitude of the net force experienced by the toy rocket. [2]
 - Determine the magnitude of the drag force acting on the rocket at terminal velocity. [1]
 - Explain why the actual magnitude of the drag force acting on the rocket at terminal velocity is found to be smaller than the calculated value in 2. [2]

[Total: 19]

11. [PJC/Prelims 2014/P2/2]

Fig. 11.1 shows two blocks, A and B, connected by a light, inextensible cord over a frictionless pulley. Block A of mass 2.0 kg rests on a horizontal smooth surface and is connected to a light spring of force constant 400 N m^{-1} . Block B of mass 10 kg is hung vertically by the cord. Initially, block A and block B are held at rest and the spring is at its original, unstretched position. The spring obeys Hooke's law.

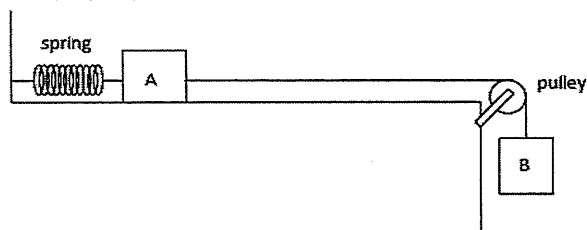


Fig. 11.1

Block B is released from rest and falls vertically to a maximum distance.

- a. Describe the energy changes that take place for the system. [4]
- b. i. Calculate the maximum distance fallen by block B. [2]
- ii. Calculate the speed of block A when block B has fallen to half of its maximum distance. [2]

[Total: 8]

12. [SRJC/Prelims 2014/P2/4]

In Fig 12.1 below, a block of mass 2.0 kg is placed against a spring on a rough plane inclined at angle 35° . The block is held in position and it compresses the spring by $x\text{ m}$. The spring has a spring constant of 1500 N m^{-1} .

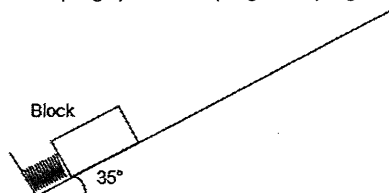


Fig. 12.1

The block is then released. It moves upwards along the slope and after separating itself from the spring, it moves up by 0.50 m along the plane before coming to a momentary stop as shown in Fig. 12.2. The average frictional force experienced by the block as it moves up the plane is 2.0 N .

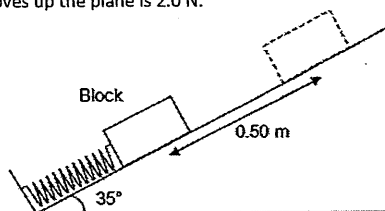


Fig. 12.2

- a. In terms of x , write down an expression for the work done by friction from the time the block is released until it comes to a momentary stop. [1]
- b. Calculate the value of x . [2]
- c. In the axes in Fig. 12.3, sketch the following graphs from the time the block separates itself from the spring until it reaches a momentary stop.

The distance represents the distance along the plane.

- i. work done against friction (WD) against distance [1]
- ii. total mechanical energy (TME) of block against distance [1]
- iii. gravitational potential energy (GPE) against distance [1]
- iv. kinetic energy (KE) against distance [1]

Label each of the graphs as WD, TME, GPE and KE respectively.

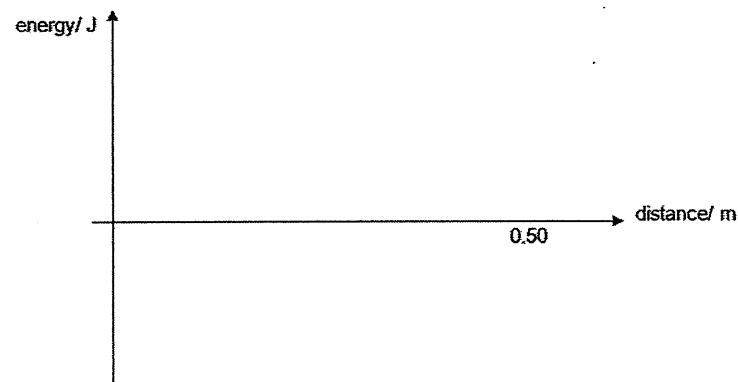


Fig. 12.3

[Total: 7]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	D	D	C	C	C	D	D	C	A	D	C	D	C	C

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	B	D	C	A	B	A	B	B	D	C	C	B	A	A

31	32
D	D

1 B See definitions in lecture notes.

2 D Assuming that drag is proportional to v
 Net force acting on sphere is $F_{\text{net}} = mg - kv$
 $ma = mg - kv$
 $-kv = ma - mg$

Differentiating the above equation with respect to time t ,

$$-k \frac{dv}{dt} = m \frac{da}{dt} \left[\frac{dv}{dt} = a \right]$$

$$-ka = m \frac{da}{dt}$$

$$\frac{da}{dt} = -\frac{ka}{m}$$

At time $t = 0$, sphere is just released from rest and $v = 0$, $a = g$.

$a = 0$, is when sphere reaches terminal velocity (where v is a maximum constant value).

Note that $\frac{da}{dt}$ represents the gradient of the a - t graph. The negative sign indicates that the graph is downward sloping. The value of gradient $\frac{da}{dt}$ decreases with time since a decreases from g to 0. Hence a - t graph is a curve whose gradient is negative and decreases with time.

Raffles Institution

Revision Package (H2 Physics)

S – Projectile Motion

3 D Due to air resistance, the speed of ball will increase at a decreasing rate until it reaches terminal velocity. It is because the resistive force due to air is dependent on speed of object. When the speed increases, the resistive force increases. The acceleration of the object (gradient of v - t graph) will decrease until its magnitude is equal to the weight of the object and the object will reach terminal velocity (constant gradient).

4 C $v = \sqrt{v_x^2 + v_y^2} = \sqrt{1.4^2 + 1.8^2} = 2.3 \text{ m s}^{-1}$

5 C $t = \frac{s_x}{v_x} = \frac{100}{200} = 0.50 \text{ s}$
 $s_y = u_y t + \frac{1}{2} a t^2 = 0 + \frac{1}{2} (9.81) (0.50)^2 = 1.23 \text{ m}$

6 C The initial horizontal speed of the projectile is $u \cos 30^\circ = 0.866u$
 Its initial kinetic energy E is equal to $\frac{1}{2} m u^2$
 At the highest point of its flight, the speed of the projectile is equal to $0.866u$ and it moves completely horizontally. This is because the vertical speed is zero at this point, leaving only the constant horizontal speed of $0.866u$.
 The kinetic energy at this highest point $= \frac{1}{2} m (0.866u)^2 = 0.75 (\frac{1}{2} m u^2) = 0.75 E$

7 D Vertical acceleration is equal to gravitational acceleration which is constant.

8 D Resistive force due to air is dependent on speed of object. When the speed increases, the resistive force increases. The acceleration of the object will decrease until it reaches zero i.e. its magnitude is equal to the weight of the object and the ball will reach terminal velocity.

9 C $v_y = u_y + at = 20 + (9.81)(2.0) = 39.62 \text{ m s}^{-1}$
 $v = \sqrt{v_x^2 + v_y^2} = \sqrt{15.0^2 + 39.62^2} = 42.4 \text{ m s}^{-1}$

10 A The kinetic energy of the ball at the highest point is non-zero, hence options B and D are wrong.
 The vertical displacement of the ball is a parabolic equation of the form $s_y = -\left(\frac{g}{2u_x^2}\right) s_x^2 + \left(\frac{u_y}{u_x}\right) s_x$. Since the gravitational potential energy is mgs_y , the graph should be a parabolic curve as well.

11 D The ball will reach terminal velocity due to air resistance hence the gradient of the graph should approach a constant value as shown in option C and D.
 The distance travelled by the ball (area under v - t graph in the positive region) when it is going up should be equal to the distance travelled (area under v - t graph in the negative region) when it is going down. Option D shows that both the areas are approximately the same.

- 12 C By the principle of conservation of energy, the gain in kinetic energy of the balls come from the loss in their gravitational potential energy i.e.

$$\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh}$$

Hence, both balls will reach at the same speed as speed is independent of their masses.

Ball Y will have a larger kinetic energy as it has larger mass i.e. $\frac{1}{2}(2m)v^2$ compared to ball X which is $\frac{1}{2}mv^2$

13 D $s_y = u_y t + \frac{1}{2}at^2 \Rightarrow h = 0 + \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}}$

$$s_x = u_x t \Rightarrow d = v \left(\sqrt{\frac{2h}{g}} \right)$$

- 14 C After 3 s, distance fallen by flyer is $100 \sin 30^\circ \times 3 = 150$ m (acceleration is zero since speed of wingsuit flyer is constant)

Using $s_y = u_y t + \frac{1}{2}at^2$ and taking downward as positive,

$$300 - 150 = (100 \sin 30^\circ)t + \frac{1}{2}(9.81)(t^2) \Rightarrow t = 2.42 \text{ s}$$

- 15 C Using $s = \frac{1}{2}gt^2$, $t = 4.52$ s.

Within 4.52 s, relative distance covered = $(35 - 15) \times 4.52 = 90.4$ m.

So relative distance to cover before dropping bomb = $550 - 90.4 = 459.6$ m

Time to wait before releasing bomb = $459.6 / 20 = 23$ s

- 16 D At maximum height, $v_y = 0$ while $v_x = v \cos \theta$

$$\therefore \text{K.E. at maximum height} = m(v \cos \theta)^2 / 2 = \text{K.E.}_{\text{initial}} \cos^2 \theta = 15 \cos^2 30^\circ = 11 \text{ J}$$

- 17 B $v_y^2 = u_y^2 + 2as \Rightarrow v_y = \sqrt{2gh}$

$$\tan \theta = \frac{v_y}{v_x} = \frac{\sqrt{2gh}}{v}$$

Option B gives the largest θ .

- 18 D Relative speed = $70 + 20 = 90 \text{ m s}^{-1}$

$$s_y = u_y t + \frac{1}{2}at^2 \Rightarrow 500 = 0 + \frac{1}{2}(9.81)t^2 \Rightarrow t = 10.10 \text{ s}$$

$$s_x = u_x t = (90)(10.10) = 909 \text{ m}$$

- 19 C $P_{\text{out}} = 0.6 \times P_{\text{in}}$

$$\frac{V \rho g \Delta h}{t} = 0.6 \times P_{\text{in}}$$

$$P_{\text{in}} = 13.6 \text{ W}$$

- 20 A Lost in G.P.E = Gain in K.E + Work against friction

$$1(1)(9.81) = \text{K.E} + 0.5(10)$$

$$\text{K.E} = 4.8 \text{ J}$$

- 21 B When the depths of water in both vessels are equal, the height of water in each vessel is $h/2$.

The CG of the mass of water in X would have fallen through a distance of $h/4$.

Hence decrease in potential energy of the water in X = $mg(h/4) = mgh/4$

- 22 A Since body is moving at constant speed,

$$100 = f + mg \sin \theta$$

$$f = 100 - mg \sin \theta$$

$$\text{Work done by friction, } fd = (100 - mg \sin \theta)d = \left(100 - 20.0(9.81) \left(\frac{12}{40} \right) \right) \times 40 = 1645.6 = 1.6 \text{ kJ}$$

OR

$$Fd = mgh + fd$$

$$fd = Fd - mgh = 100(40) - 20.0(9.81)(12) = 1645.6 = 1.6 \text{ kJ}$$

- 23 B KE before striking the surface = (GPE before releasing the ball) which is proportional to a height of 72 cm = 0.75 J

KE just after leaving surface = (max GPE after bounce) which is proportional to a height of 37 cm

Hence, by proportion of height, KE just after leaving surface

$$= (\text{max GPE after bounce}) / (\text{GPE before releasing}) \times 0.75 \text{ J} = (37/72) \times 0.75 = 0.385 \text{ J}$$

- 24 B Net loss in G.P.E = Net gain in K.E

Loss of G.P.E of X – gain in G.P.E of Y = Gain in K.E by system

$$m_x g(2 \sin 30^\circ) - m_y g(2) = \text{Gain in K.E by system}$$

$$\text{Gain in K.E by system} = 78.48 - 58.86 = 19.6 \text{ J}$$

- 25 D Effective power = total work done / time taken = $\frac{\rho Xgh + \frac{1}{2}\rho Xv^2}{t} = \frac{2\rho Xgh + \rho Xv^2}{2t}$

- 26 C KE of particle when it just enters liquid column $KE_i = mgh$

decrease in GPE = increase in KE + work done against resistive force

$$mg(0.60) = (KE_f - KE_i) + f(0.60)$$

$$(0.80)(9.81)(0.60) = \left(\frac{1}{2}(0.80)(7.5)^2 - (0.80)(9.81)h \right) + 120(0.60)$$

$$h = 11.4 \text{ m}$$

- 27 C $\frac{mgh}{t} \times \frac{100}{80} = \frac{(400)(9.81)(1200)}{2.0 \times 60} \times \frac{100}{80} = 49050 = 49 \text{ kW}$

- 28 B At the bottom of the ramp, gain in KE of a block is equal to loss in GPE of block
 $\frac{1}{2}mv^2 = mgh$

$$v^2 = 2gh$$

$$v = \sqrt{2gh}$$

$$v \propto \sqrt{h}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{X \sin 60^\circ}{Y \sin 30^\circ}} = \sqrt{\frac{2 \sin 60^\circ}{\sin 30^\circ}} = 1.9$$

- 29 A By Principle of conservation of energy,
 Loss in GPE = Gain in KE + Work done against motion

$$2g(3) = \frac{1}{2}(2)v^2 + 4(5) \Rightarrow v = 6.2 \text{ m s}^{-1}$$

- 30 A (KE+GPE) at lowest point = (KE+GPE) at highest point

$$\frac{1}{2}m(3.0)^2 = mgh \Rightarrow h = 0.4587 \text{ m}$$

$$\cos \theta = \frac{4.0 - 0.4587}{4.0} \Rightarrow \theta = 27.7^\circ$$

- 31 D W.D by person = Gain in GPE + W.D against friction
 $= mgh + fd = 200(1.5) + 150(1.5 / \sin 30^\circ)$
 $= 750 \text{ J}$

- 32 D power output = $\frac{mgh}{t} = \frac{10 \times 1.00}{5.0} = 2.0 \text{ W}$

$$0.25 = \frac{2.0}{P_{\text{supplied}}} \Rightarrow P_{\text{supplied}} = 8.0 \text{ W}$$

Structured Questions

1. [ACJC/Prelims 2014/P2/7]

- a. See definitions in notes.
- b. i. distance is area under graph, suitable method to find area, range accepted: 380 – 400 m
- ii. First part: Uniform acceleration, only weight/gravitational force acting
 Second part: Acceleration decreasing / speed increasing at decreasing rate, Air resistance increases as speed increase hence resultant force downwards decreases
 Third part: Uniform speed, Air resistance = weight / resultant force zero

- iii. tangent drawn with tangent drawn at 6.0 s, 2.84 m s^{-2}

t / s	v / m s ⁻¹	a / m s ⁻²	(g - a) / m s ⁻¹	(g - a) / v ² / 10 ⁻³ m ⁻¹
3.0	28.5	6.90	3.10	3.81
6.0	43.2	2.84	7.16	3.84
9.0	48.0	1.20	8.80	3.82

(g - a) at 6.0 s determined correctly

Resultant force = $mg - kv^2 = ma$

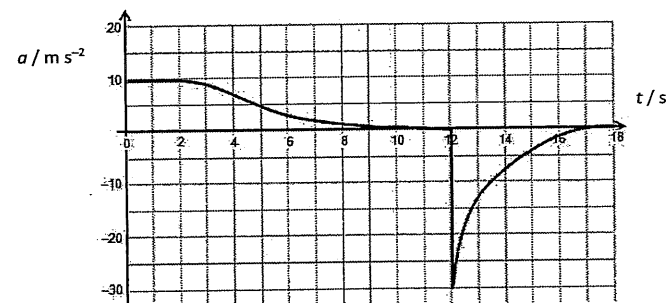
Hence if F_v is proportional to v^2 , $(g - a)/v^2$ is a constant.

Calculated value of $(g - a)/v^2$ all correct

Since values are almost the same, hence true.

- c. i. tangent drawn and gradient of tangent calculated to show that $a = -30 \text{ m s}^{-2}$

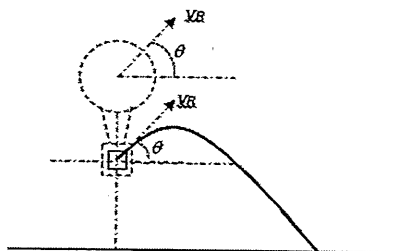
ii.



2. [AJC/Prelims 2014/P2/1]

- a. Constant acceleration and motion in a straight line
- b. i. $v_R = \sqrt{v_x^2 + v_y^2} = \sqrt{10.0^2 + 3.0^2} = 10.4 \text{ m s}^{-1}$
 $\theta = \tan^{-1}\left(\frac{10}{3}\right) = 73.3^\circ$

ii. 1.



2. Taking downwards as positive,

$$s_{\text{bag}} = -10.0 (4.0) + \frac{1}{2} (9.81) (4.0)^2 = 38.48 \text{ m}$$

$$s_{\text{balloon}} = -10.0 (4.0) = -40.0 \text{ m}$$

$$\text{Distance apart} = 40.0 + 38.48 = 78.5 \text{ m}$$

3. [JC/Prelims 2014/P2/1(part)]

a. i. Taking upwards as positive,

$$\text{Using } v_y^2 = u_y^2 + 2as \Rightarrow 0 = u_y^2 + 2(-9.81)(3.2 - 1.7) \Rightarrow u_y = 5.4 \text{ m s}^{-1}$$

ii. Applying Newton's 2nd law of motion, and taking rightwards as positive

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t} = \frac{m(v - u)}{t} = \frac{0.024(-12 - 15)}{0.15} = -4.32 \text{ N}$$

(Force on ball by wall which acts to the left)

By Newton's 3rd law of motion,

Force on wall by ball = - Force on ball by wall = 4.32 N

Force on wall acts to the right.

iii. It is not elastic as the speed of the ball has decreased from 15 m s^{-1} to 12 m s^{-1} which means that the KE of the ball-wall system, which is $\frac{1}{2}mv^2$, has decreased.

iv. As the horizontal velocity is reduced after collision, with the same time of flight before collision with the wall, the horizontal displacement during rebound will be reduced. (The ball will land closer to the wall.)

Students must mention the reduction in both horizontal velocity and displacement.

4. [JC/Prelims 2014/P2/1]

a. The ramp cannot be frictionless so that the ramp can exert a forward force on the driving wheel (s) to move the motorcycle up the ramp.

b. When he takes off and reaches the same height as the car, $s_y = 0 \text{ m}$

$$s_y = u_y t + \frac{1}{2} a t^2$$

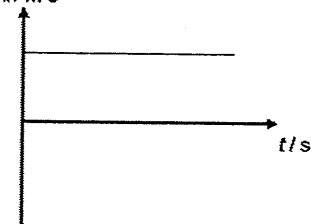
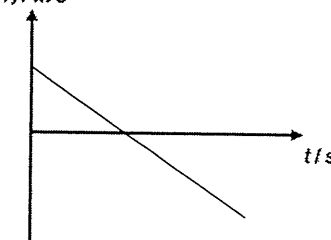
$$0 = \left(14 \sin 30^\circ + \frac{1}{2} (-9.81) t \right) t$$

$$t = 0 \text{ s OR } 1.427 \text{ s}$$

Let x = number of cars that can jump over

$$1.6x = (14 \cos 30^\circ)(1.427) = 17.3$$

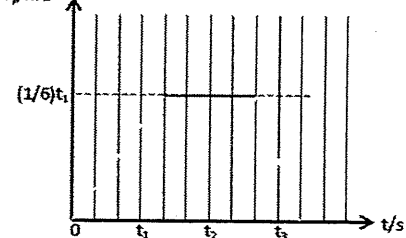
$$x = 10$$

c. i. $v_x / \text{m s}^{-1}$ ii. $v_y / \text{m s}^{-1}$ 

5. [JC/Prelims 2014/P2/2]

a. Constant acceleration and motion in a straight line

b. i. The passengers have gone to a higher floor. Positive acceleration points in the upwards direction. Area under the a-t graph gives the change in velocity. The two areas under the graph are equal and opposite in value. Thus, lift moves upwards initially, increasing in speed. It then travels in constant velocity for some time before slowing down and stopping.

ii. $v_y / \text{m s}^{-1}$ 

c. i. Taking a path similar to path 2,

$$\text{Time of flight} = 2V_o \sin \alpha / g,$$

where V_o is initial speed and α is the angle V_o make with horizontal

$$\text{Horizontal Range} = \text{Horizontal velocity} \times \text{time of flight} = V_o^2 \sin 2\alpha / g$$

$$\text{Hence, } V_o^2 \sin 90^\circ / g = V_o^2 \sin 2\theta / g + \frac{1}{4} V_o^2 \sin 2\theta / g$$

$$\theta = 26.6^\circ$$

ii. $[2V_o \sin 26.6^\circ / g + V_o \sin 26.6^\circ / g] / 2V_o \sin 45^\circ / g = 0.949$

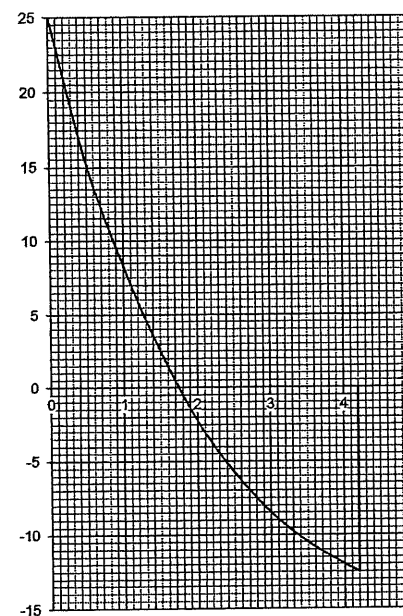
6. [NYJC/Prelims 2014/P2/1]

- a. Mass of ball = 0.050 kg
 Floor-to-ceiling height = 3 m
 Height at the top of building = $12 \times 3 = 36$ m
 Using $v^2 = u^2 + 2as$, velocity before ball reaches ground = 26.6 m s^{-1}
 Momentum = $mv = 1.3 \text{ N s}$
- b. i. $t = 0$ to 0.1 s : $s = 0.007 = \frac{1}{2} a (0.1)^2$
 $t = 0.1$ to 0.2 s : $s = 0.029 = \frac{1}{2} a (0.1)^2$
 $t = 0.2$ to 0.3 s : $s = 0.065 = \frac{1}{2} a (0.1)^2$
 $t = 0.3$ to 0.4 s : $s = 0.116 = \frac{1}{2} a (0.1)^2$
 Plot s against t^2
- ii. Gradient = $(0.016 - 0.094) / (0.02 - 0.13) = 0.70 \text{ m s}^{-2}$
 Acceleration = $2 \times \text{gradient} = 1.4 \text{ m s}^{-2}$
- c. i. $s = ut + \frac{1}{2} at^2$
 $30 = (15 \sin 55^\circ)t - \frac{1}{2} (1.2t^2)$
 $t = 2.83 \text{ s}$ or 17.6 s (NA)
- ii. $s = ut = 15 \cos 55^\circ (2.83) = 24.4 \text{ m}$
- iii. The acceleration of free fall on Planet Newtonia (1.4 m s^{-2}) is larger than that on the moon. It would take longer for the projectile to reach a height of 30 m.

7. [PJC/Prelims 2014/P2/1]

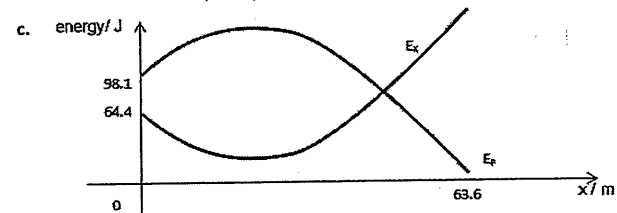
- a. The gradient of the graph represents the acceleration, which is changing with speed.
 This suggests that the resultant force acting on the ball is changing with speed, and since weight is constant, it must have meant that air resistance changes with speed.
- b. For the upward motion,
 loss in KE = gain in GPE + energy lost
 $\frac{1}{2} \times 0.350 \times 25^2 = (0.350 \times 9.81 \times 19) + \text{energy lost}$
 energy lost = 44.14 J
 For the downward motion,
 loss in GPE = gain in KE + energy lost
 $0.350 \times 9.81 \times 19 = \left(\frac{1}{2} \times 0.350 \times 12.5^2 \right) + \text{energy lost}$
 energy lost = 37.89 J
 Therefore, ratio = $\frac{44.14}{37.89} = 1.16$
- c. i. 1.75 s

ii.



8. [SAJC/Prelims 2014/P3/1]

- a. At maximum height, $v_y = 0$
 Taking upwards as positive, using $v_y^2 = u_y^2 + 2as_y$
 $0 = (20 \cos 30^\circ + 6)^2 + 2(-9.81)s_y$
 $s_y = 27.7 \text{ m}$
 Max height above ground = $27.7 + 50 = 77.7 \text{ m}$
- b. Taking upwards as positive, $s_y = u_y t + \frac{1}{2} a t^2$
 $-50 = (20 \cos 30^\circ + 6)t + \frac{1}{2}(-9.81)t^2$
 $4.905 t^2 - 23.3205 t - 50 = 0$
 $t = \frac{23.3205 \pm \sqrt{(23.3205)^2 - 4(4.905)(-50)}}{2(4.905)} = \frac{23.3205 \pm \sqrt{1524.84572}}{2(4.905)} = 6.36 \text{ s}$



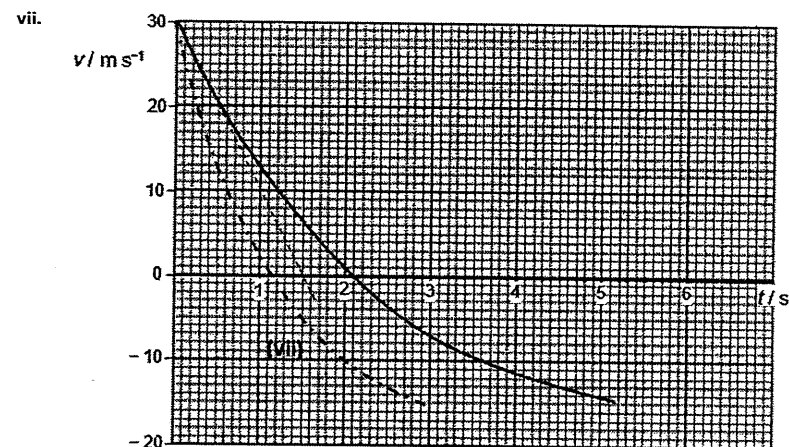
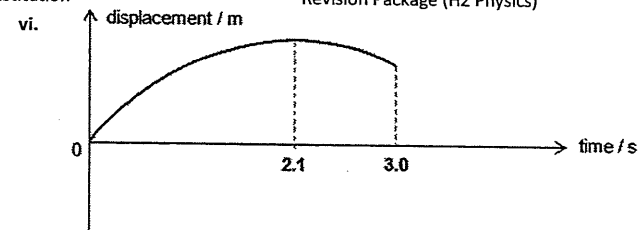
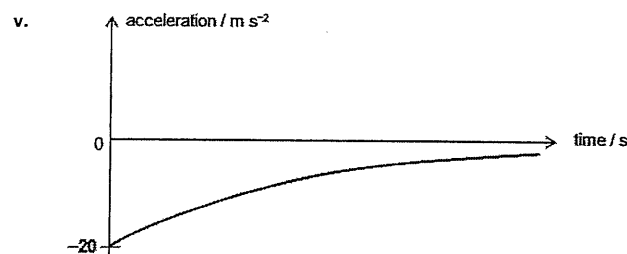
- d. Air resistance causes shorter range (maximum horizontal distance) and lower maximum height. Since it opposes motion in both x and y directions.

9. [SRJC/Prelims 2014/P2/1]

- a. i. Horizontal component of velocity = 8.2 m s^{-1}
Vertical component of velocity = $8.2 \tan 60^\circ = 14.2 \text{ m s}^{-1}$
- ii. Using $v = u + at$, time of descent = $14.2 / 9.81 = 1.45 \text{ s}$
 $x = 1.45 \times 8.2 = 11.9 \text{ m}$
- b. i. When the stone first enters the sea with a high speed, it will experience a high amount of resistive force. The resistive force will be larger than weight, hence having an upwards resultant force. Hence, it will experience an upward acceleration which will cause the stone to slow down as it falls downwards.
As the speed decreases, the resistive force will decrease accordingly and hence causing the magnitude of the acceleration to decrease.
With that, the rate of the stone slowing down its speed will decrease till it finally attains terminal velocity.
- ii. It will reach a higher terminal velocity.

10. [YJC/Prelims 2014/P3/6]

- a. See definitions in lecture notes.
- b. i. Since the gradient of the graph decreases as speed decreases, the drag force is proportional to the speed of the rocket.
Drag force increases when the rocket is propelled upwards as speed increases.
Since resultant force when the rocket is propelled upwards is constant, propulsion force must increase.
- ii. $a = \text{gradient of tangent drawn} = (30 - 0) / 1.4 = 21.4 \text{ m s}^{-2}$
- iii. There is drag force due to air resistance acting downwards when the rocket is moving upwards.
Resultant force is greater than the weight.
- iv. Time is 2.1 s.
Velocity is zero, therefore no drag force acting on the rocket. Resultant force equals to weight thus resultant acceleration equals to g .



- viii. 1. Average $a = (v - u) / t = (3 - 22) / 3 \times 10^{-3} = -6300 \text{ m s}^{-2}$
 $F_{\text{net}} = ma = 2.0 \times 6300 = 12600 \text{ N}$
2. $F_D = 2.0 \times 9.81 = 19.62 \text{ N}$
3. There is significant upthrust acting on the rocket.
The weight is balanced off by the sum of upthrust and the drag force at terminal velocity.

11. [PJC/Prelims 2014/P2/2]

- a. Initially, block B has maximum gravitational potential energy.
When block B falls, it loses gravitational potential energy. Both block A and B gain kinetic energy. The spring stretches beyond its original length and there is an increase in elastic potential energy in the spring.
As the spring gains elastic potential energy and stretches the system beyond its equilibrium position, there is loss in kinetic energy of the blocks.
The blocks come to a rest momentarily when block B has fallen to a maximum distance. At this position, the elastic potential energy in the spring is equal to the total loss in gravitational potential energy of block B. The kinetic energy of the blocks is zero.

- b. i. When block B has fallen to its maximum distance, it comes to rest momentarily.

Let the maximum distance fallen be h .

Loss in GPE of block B = Gain in EPE in spring

$$mgh = \frac{1}{2} kx^2$$

$$mgh = \frac{1}{2} kh^2$$

$$h = \frac{2mg}{k} = \frac{2(10)(9.81)}{400} = 0.491 \text{ m}$$

- ii. Half of the maximum distance = $\frac{1}{2}(0.491) = 0.245 \text{ m}$

loss in GPE of block B = gain in EPE of spring + gain in KE of blocks A and B

$$mgh' = \frac{1}{2} kx^2 + \frac{1}{2} (M + m)v^2$$

$$\frac{1}{2} (2 + 10)v^2 = 10(9.81)(0.245) - \frac{1}{2} (400)(0.245)^2$$

$$v = 1.42 \text{ m s}^{-1}$$

12. [SRJC/Prelims 2014/P2/4]

a. $W_{\text{by friction}} = -2(x + 0.5) = -(2x + 1)$

b. initial total energy = final total energy

$$\text{EPE} = W_{\text{against friction}} + \text{GPE}$$

$$\frac{1}{2} (1500) (x^2) = 2(x + 0.5) + 2(9.81)(x + 0.5) \sin 35^\circ$$

$$x = 0.10 \text{ or } -0.09 \text{ (NA)}$$

